

Mech Recruitment Project



By Zoe Lahanas

Break down

01 **About me**

Slide 03

02 **Materials**

Slide 06

03 **Chassis Design**

Slide 07

About me

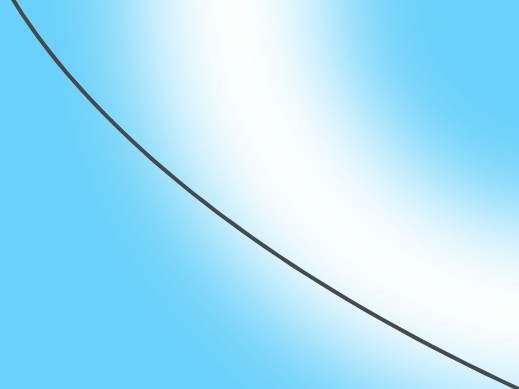


Zoe Lahanas

I'm a sophomore from Walnut Creek studying Mechanical Engineering. I love working out, hanging out with my friends, spending time outside, and trying new foods. My dream is to end up working in the F1 circuit. My goals for this year are to climb half dome and try every flavor of Ben and Jerry's ice cream.







WHY'S IT SPECIAL?

Materials



AISI 4130 Chromoly

- Weldability
 - Good strength to weight ratio
 - Not too expensive and used widely throughout the FSAE community
 - Super available
 - High yield strength
- Modulus of Elasticity = 190-210 GPa

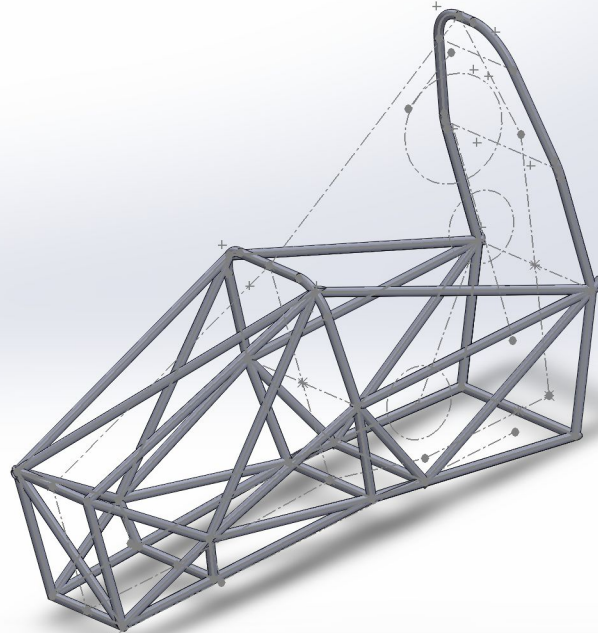


Applications in

- Aerospace
- Manufacturing
- BMX/Motorcycles

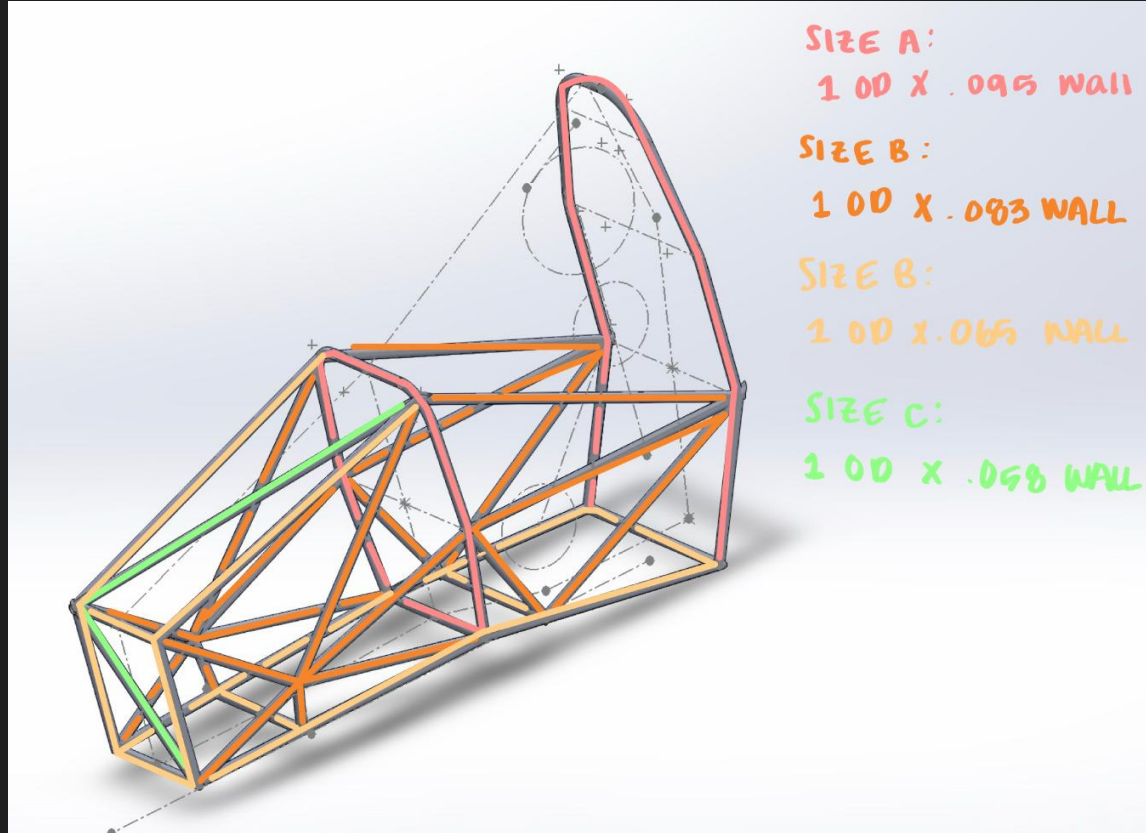
Stats

- Uses 3,435,503.43 mm³ of steel
- Weighs 26.969 kg
- Empty FSAE EV chassis weighs between 25-45 kgs which is perfect because it's not complete
- 1.541m long and 1.105m high



Tube Dimensions

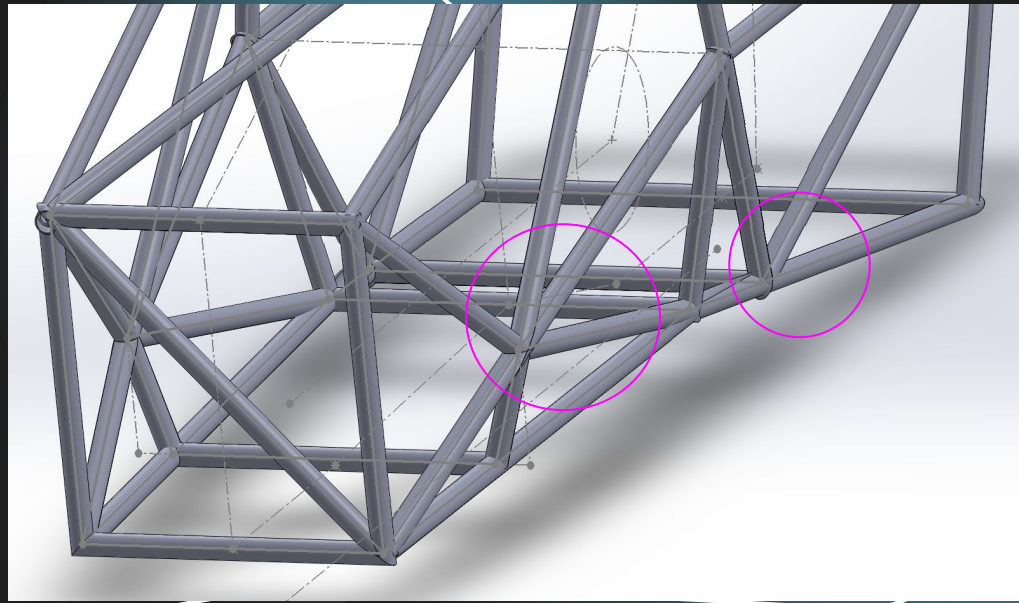
Opted for two different variations of a Size B tube for added stability

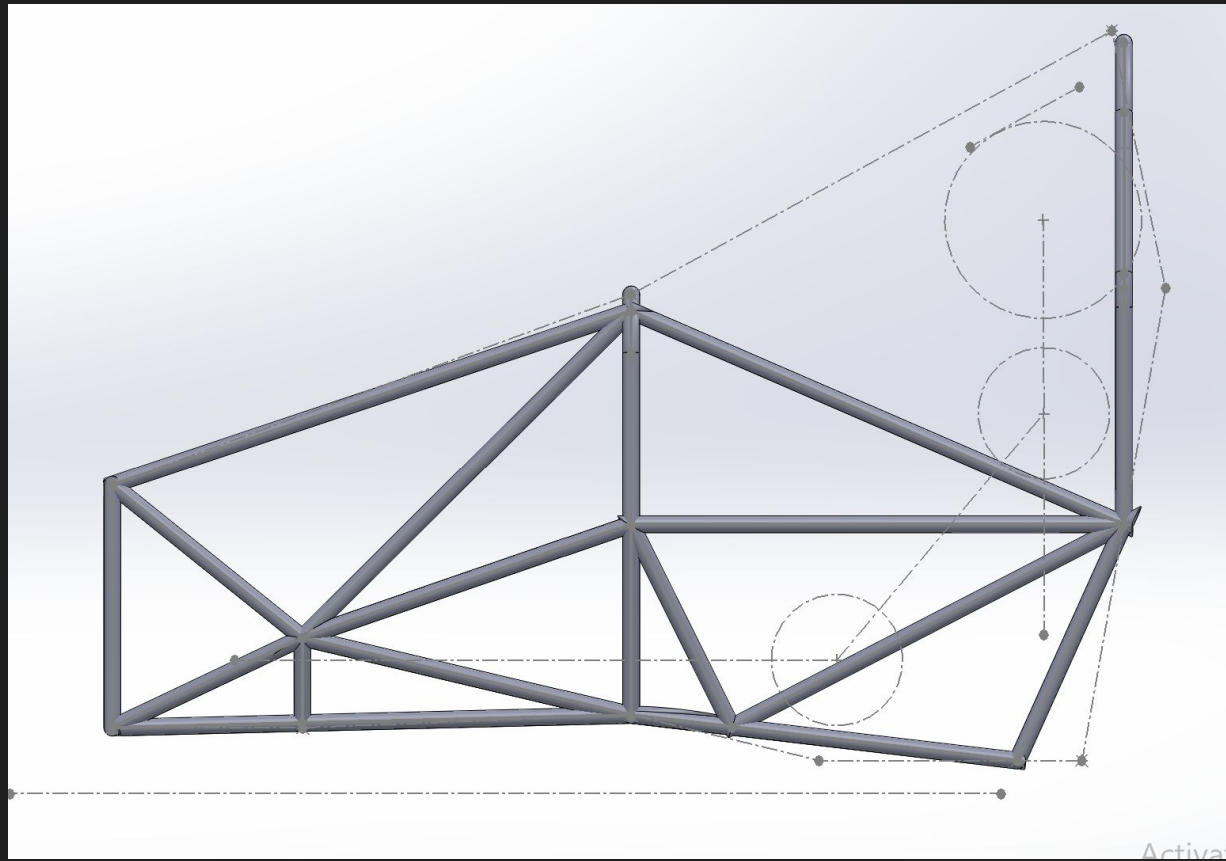


Side Impact Structures and Torsional Rigidity

Why do most cars side wall protrude out slightly?

- Designed to enhance torsional rigidity by making the chassis more box like without making too wide of a base
- It also provides more space to the driver while following the design rules for the SIS

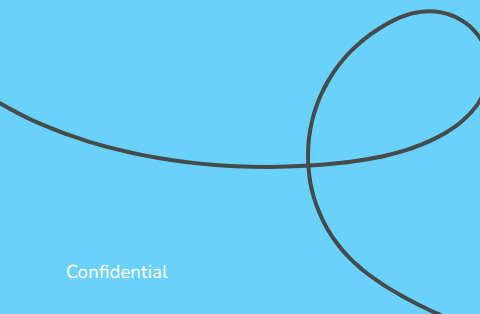
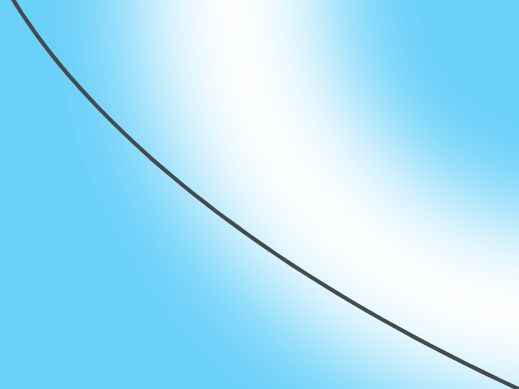




Dropped the center of gravity of the driver to enhance the cars traction on the ground and assist with turns and accelerations

My most useful dimensions were

- Foot width and length
- Erect sitting height
- Shoulder Grip Length
- Sitting Shoulder Height and Width



Questions?